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VPC Traffic Flow and Security

AL alexten3517@gmail.com

alexten3517@gmail.com
NextWork Student

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Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a flexible and customizable networking environment whenever you want to have a private cloud which will consist of several personalized resource which will only pertain to specific protocols or IP addresses.

How I used Amazon VPC in this project

I used it to practice creating a subnet,a route table and a security group which will help secure an isolated enviorment for any EC2 intance.,whether private or public.

One thing I didn't expect in this project was...

How much fun it was to create.

This project took me...

Spent about 40 mins tops.

AL

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NextWork Student

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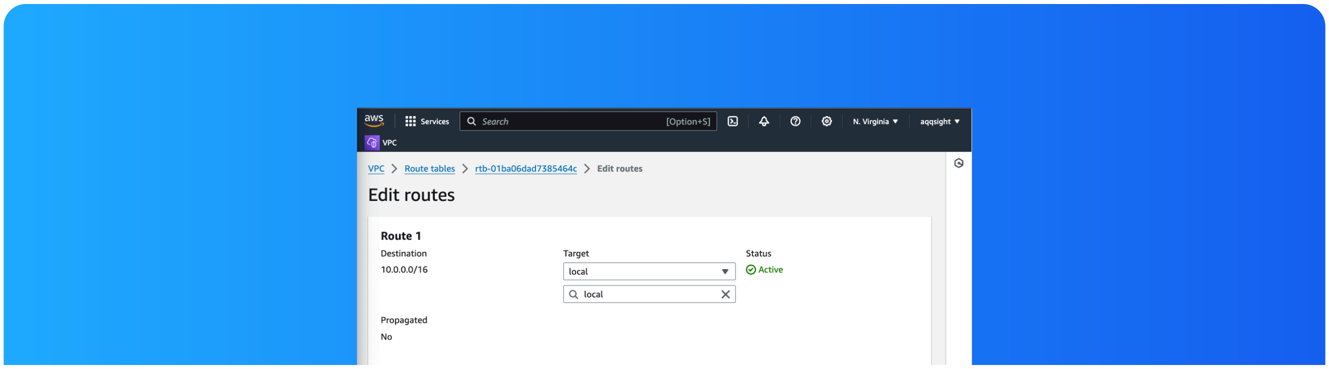
Route tables

Route tables are roadmaps that allow data packets to either flow locally(connect with local resources within the VPC) or to the IGW which allows for communication with private networks on the internet.

Routes tables are needed to make a subnet public because they help direct the traffic across networks by making sure the traffic reaches it destination.



AL
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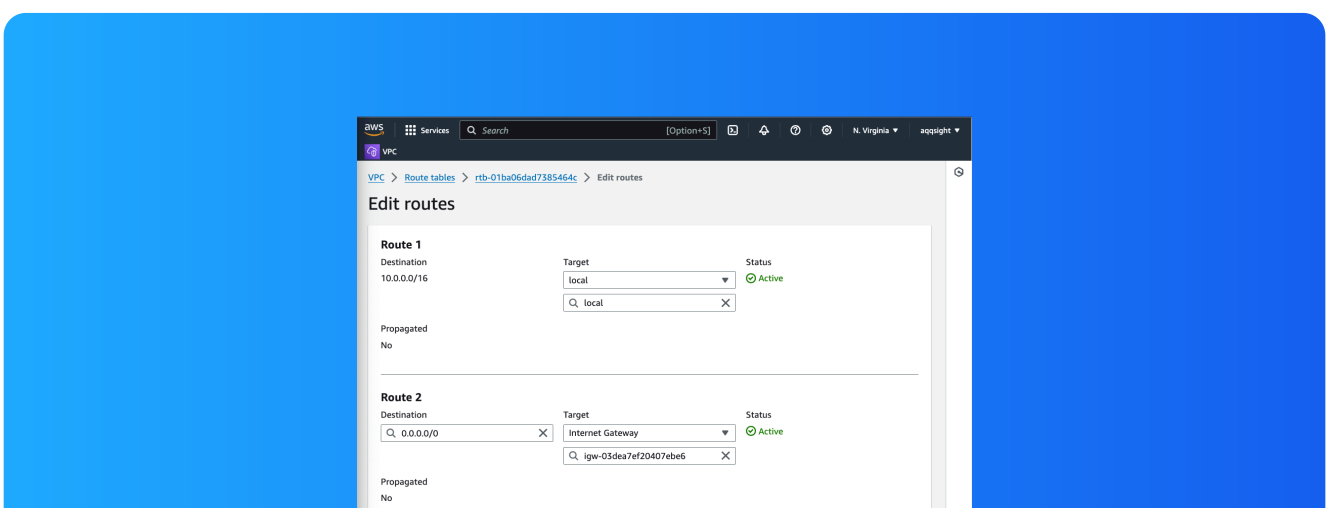
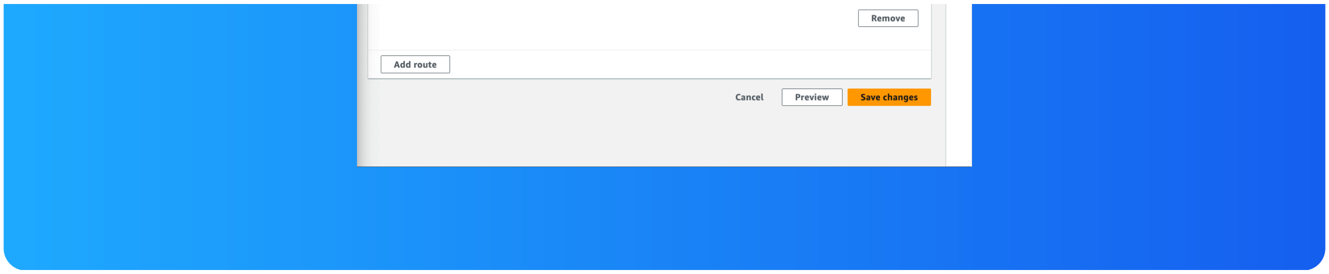
Route destination and target

Routes are defined by their destination and target, which mean they will be sent or recieved to specific IP address(s).They also indicate the interface where traffic should be sent to reach that destination(another subnet,an IGW,VPN or a instance).

The route in my route table that directed internet-bound traffic to my internet gateway had a destination of a range of IP addresses.... and a target of the path that the traffic use to get to their destination.





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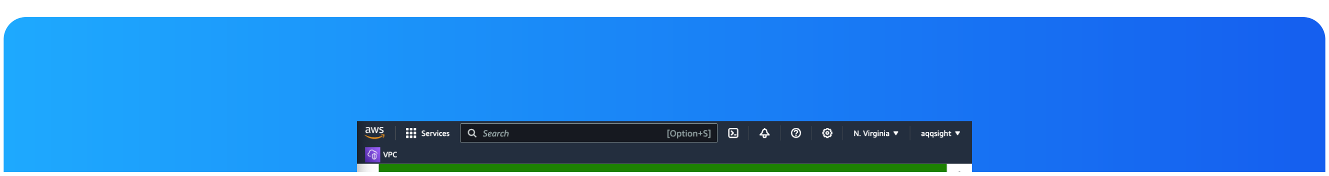
Security groups

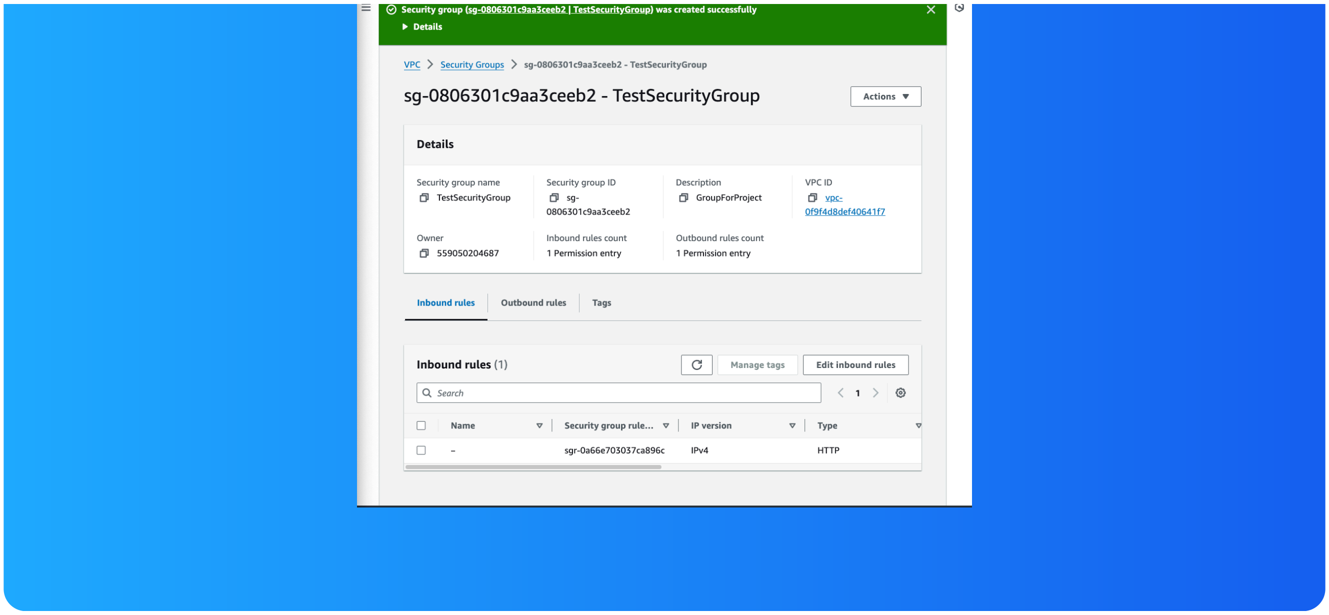
Security groups are are rules that allow specified traffic to go inbound or outboud at the resources level. Every single subnet/resource has a security group attached to it.

Inbound vs Outbound rules

Inbound rules are to allow what traffic comes into the VPC. These rules restrict/monitor all inbound HTTP traffic.

Outbound rules are set to allow all traffic to leave the VPC.. By default, my security group's outbound rule is All Traffic.





AL

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Network ACLs

Network ACLs are gate keepers for our traffic to go through our subnets or not. They secure our network at the subnet level. They maintain network accessibility for all data packets coming in and going out.

Security groups vs. network ACLs

The difference between a security group and a network ACL is that security group guard our resources while our ACL's maintain a level of security at the newtork level.

AL

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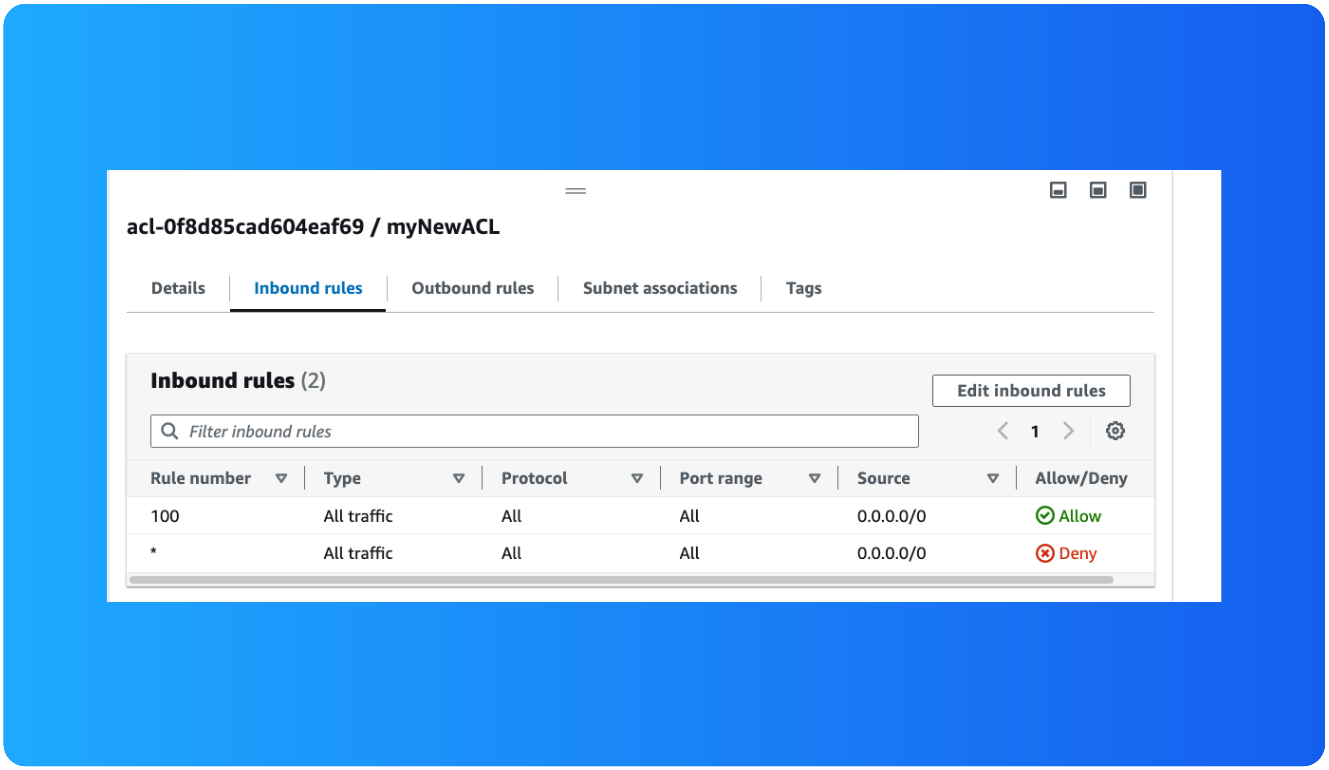
Default vs Custom Network ACLs

Similar to security groups, network ACLs use inbound and outbound rules

By default, a network ACL's inbound and outbound rules will allow all incoming traffic coming into our subnets. By default they keep a traffic coming until we create your own custom rules to restrict traffic as needed.

In contrast, a custom ACL's inbound and outbound rules are automatically set for protocols,ports and IP addresses to allow or deny traffic accordingly.



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